Avik Poudel

AIT, Pathum Thani, Thailand | poudel.avik91@gmail.com | +66-988500785 | LinkedIn

PROFILE

Structural engineer and MEng candidate in Structural Engineering at the Asian Institute of Technology (AIT), registered with the Nepal Engineering Council. Brings industry experience in structural design, site supervision and project documentation, and is currently interning at CSI Bangkok. Skilled in ETABS, SAFE, STAAD.Pro and AutoCAD, with a strong foundation in nonlinear finite element programming and structural optimization. Seeking an entry-level structural engineering role to deliver safe, economical and code-compliant designs.

EDUCATION

Asian Institute of Technology (AIT), Pathum Thani, Thailand

Master of Engineering in Structural Engineering — Expected May 2027

Relevant Coursework: Finite Element Method, Nonlinear Structural Analysis, Structural Dynamics, Prestressed Concrete, Stability of Steel Structures and Wind Engineering.

Pokhara University, Cosmos College of Management and Technology, Lalitpur, Nepal

Bachelor of Civil Engineering — September 2018 – August 2023

Cumulative **GPA:** 3.6/4.0

Thesis Title: Design and Analysis of Multistoried Semi-Commercial RCC Building

Relevant Coursework: Strength of Materials, Structural Analysis I, Structural Analysis II, Design of Steel and Timber Structures, Design of RCC Structures, Concrete Technology and Masonry Structures.

United Academy — Kathmandu, Nepal

Cumulative **GPA:** 3.1/4.0 — July 2016 – August 2018

Rhododendron International Boarding School — Kathmandu, Nepal

Cumulative **GPA:** 3.35/4.0 — April 2006 – July 2016

EXPERIENCE

Intern [CSI Bangkok, Thailand] — May 2026 – Present

Civil Engineer [Hints Consult Pvt. Ltd., Kathmandu, Nepal] — Dec 2023 – Feb 2025

(Developing project plans including schedules, budgets and resource allocation)

Major Responsibilities:

- Structural design and analysis.
- Overseeing project sites.
- Preparing and maintaining accurate project documentation (progress reports, design modifications and compliance reports).
- Engaging with clients to understand their requirements, provide updates and address any concerns, ensuring a high level of client satisfaction.

ACADEMIC PROJECTS

Nonlinear FE Analysis of a Concrete Portal Frame [AIT] 2026

- Developed a nonlinear FE program in MATLAB using Q8 elements, Gmsh meshing and a hyperbolic concrete material model; results benchmarked against ANSYS.
- Implemented incremental, Newton–Raphson and displacement-control solvers.

Structural Optimization of a 72-Bar Space Truss [AIT] 2026

- Minimised the weight of the 72-bar space truss benchmark by sizing 16 member groups under stress and nodal displacement constraints.
- Coupled a truss finite element analysis with an optimization algorithm and compared results with published benchmark solutions.

Design and Analysis of Multistoried Semi-Commercial RCC Building May 2023 – August 2023

- Led a team in the load calculation, structural layout and design of the building, including its foundation.
- Modelled and analysed the structure in ETABS and AutoCAD to NBC 105:2020 and IS 875:1987.

Surveying Camp of Jarsing Pauwa, Nepal May 2022

- 10-day camp covering topographic, road alignment and bridge site surveys using total station, theodolite and levelling machine.
- Prepared topographic and contour maps using closed traverse, triangulation and reciprocal levelling.

Minor Projects: Programming in C and C++ to solve beam problems by the conjugate beam method; manufacture and marketing of different kinds of bricks.

SKILLS

- Proficient in AutoCAD, ETABS, SAFE and STAAD.Pro
- Knowledge of programming languages such as C, C++ and HTML
- Proficient in Microsoft Office (Word, Excel, PowerPoint)
- Fluent in Nepali and Hindi; proficient in English
- Strong analytical and problem-solving skills

PROFESSIONAL MEMBERSHIP

Nepal Engineering Council, Registration No. 80477

EXTRA CURRICULAR ACTIVITIES

General Member [Civil Engineering Student Society, CESS-Cosmos] December 2022 – August 2023

- Participated in various competitions both within and outside the college.
- Attended seminars related to civil engineering innovations.

LEADERSHIP, GRANT, AWARD AND HONOR

Leadership: Leader in all major and minor academic projects; Best Leadership amongst 24 groups during Survey Field Project

Grant and Scholarship: Cosmos College of Management and Technology, Rs. 400,000 2018 – 2023

Awards and Honors: Cosmos College of Management and Technology: *Semester Topper, 7^{th} semester*